10TH CLASS MATHEMATICS — NEW SYLLABUS (PECTAA)

Exercise 6.1 — Vectors in the Plane

Unit 6 | Solved Exercise | New Book (PECTAA)

Exercise 6.1 (Solved)

A vector in the plane is written as xi + yj. Its magnitude is $|a| = \sqrt{(x^2 + y^2)}$, and the unit vector in its direction is a / |a|. Two vectors are parallel when one is a scalar multiple of the other.

1. Name the quadrant in which each point lies.

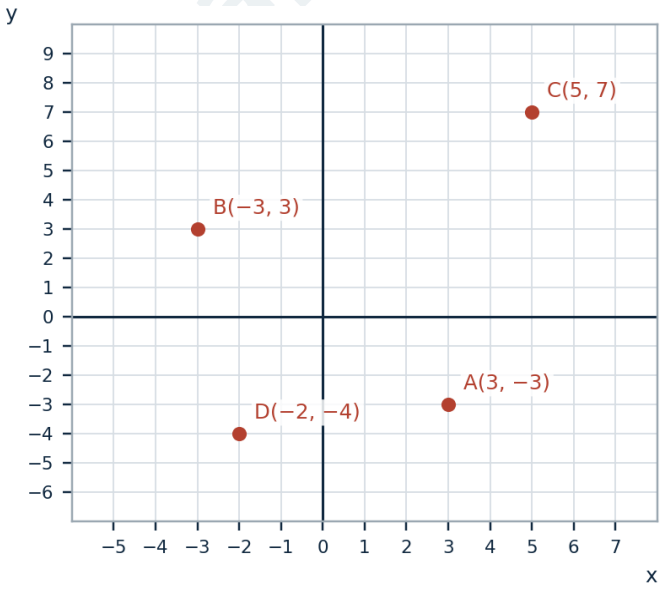
Quadrant of each point

Point	Signs	Quadrant
(i) (4, 3)	(+, +)	First
(ii) (5, −4)	(+, −)	Fourth
(iii) (−6, 2)	(−, +)	Second
(iv) (−4, −4)	(−, −)	Third

In the first quadrant both numbers are positive; in the second x is negative and y is positive; in the third both are negative; in the fourth x is positive

2. Plot the following points on the coordinate plane.

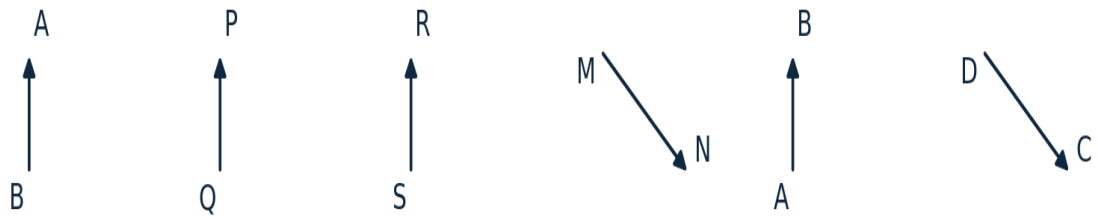
(i) A(3, −3) (ii) B(−3, 3) (iii) C(5, 7) (iv) D(−2, −4)



All four points plotted on the plane

Answer: A lies in the fourth quadrant, B in the second, C in the first, D in the third

3. Name the tail and the tip of the following vectors.



The arrow head shows the tip of each vector

Tail and tip

Vector	Tail	Tip
(i)	B	A
(ii)	Q	P
(iii)	S	R
(iv)	M	N
(v)	A	B
(vi)	D	C

The tail is the starting point of the arrow, the tip is the end point.

4. Write the vector AB in the form xi + yj.

$\overline{AB} = (x_2 - x_1)\bar{i} + (y_2 - y_1)\bar{j}$

(i) A(1, −7), B(−2, 4)

$\overline{AB} = (-2 - 1)\bar{i} + (4 - (-7))\bar{j}$

$= -3\bar{i} + 11\bar{j}$

Answer: $\overline{AB} = -3\bar{i} + 11\bar{j}$

(ii) A(8, 9), B(12, 3)

$\overline{AB} = (12 - 8)\bar{i} + (3 - 9)\bar{j}$

$= 4\bar{i} - 6\bar{j}$

Answer: $\overline{AB} = 4\bar{i} - 6\bar{j}$

5. Find the magnitude of a.

$|a| = \sqrt{(x^2 + y^2)}$

(i) a = −3i + 2j

$|a| = \sqrt{((-3)^2 + (2)^2)}$

$= \sqrt{(9 + 4)}$

Answer: $|a| = \sqrt{13}$

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(ii) $a = 4i - 3j$

$|a| = \sqrt{((4)^2 + (-3)^2)}$

$= \sqrt{(16 + 9)}$

Answer: $|a| = 5$

(iii) $a = 1/2i + 3/2j$

$|a| = \sqrt{(\frac{1}{4} + \frac{9}{4})}$

$= \sqrt{(\frac{10}{4})}$

Answer: $|a| = \sqrt{10}/2$

6. Find a unit vector in the direction of the vector given below.

Unit vector $= a / |a|$

$= (x\bar{i} + y\bar{j}) / \sqrt{(x^2 + y^2)}$

(i) $a = -4i + 5j$

$|a| = \sqrt{(16 + 25)} = \sqrt{41}$

$\hat{a} = \frac{-4}{\sqrt{41}}\bar{i} + \frac{5}{\sqrt{41}}\bar{j}$

Answer: $\hat{a} = (-4/\sqrt{41})i + (5/\sqrt{41})j$

(ii) $a = 6i + 8j$

$|a| = \sqrt{(36 + 64)} = \sqrt{100} = 10$

$\hat{a} = \frac{6}{10}\bar{i} + \frac{8}{10}\bar{j}$

Answer: $\hat{a} = (3/5)i + (4/5)j$

(iii) $a = 1/\sqrt{6}i + 1/\sqrt{6}j$

$|a| = \sqrt{(\frac{1}{6} + \frac{1}{6})} = \sqrt{(\frac{1}{3})} = \frac{1}{\sqrt{3}}$

$\hat{a} = \sqrt{3} \cdot \frac{1}{\sqrt{6}}\bar{i} + \sqrt{3} \cdot \frac{1}{\sqrt{6}}\bar{j}$

Answer: $\hat{a} = (1/\sqrt{2})i + (1/\sqrt{2})j$

(iv) $a = 1/2i - 3/4j$

$|a| = \sqrt{(\frac{1}{4} + \frac{9}{16})} = \sqrt{(\frac{13}{16})} = \frac{\sqrt{13}}{4}$

$\hat{a} = \frac{1}{2} \times \frac{4}{\sqrt{13}}\bar{i} - \frac{3}{4} \times \frac{4}{\sqrt{13}}\bar{j}$

$= \frac{2}{\sqrt{13}}\bar{i} - \frac{3}{\sqrt{13}}\bar{j}$

Answer: $\hat{a} = (2/\sqrt{13})i - (3/\sqrt{13})j$

7. If $a = 5i - 7j$, $b = -i - j$ and $c = 2i + 3j$, then find a unit vector parallel to $a + b - 3c$.

$a + b - 3c = (5\bar{i} - 7\bar{j}) + (-\bar{i} - \bar{j}) - 3(2\bar{i} + 3\bar{j})$

$= (5 - 1 - 6)\bar{i} + (-7 - 1 - 9)\bar{j}$

$= -2\bar{i} - 17\bar{j}$

$|a + b - 3c| = \sqrt{((-2)^2 + (-17)^2)} = \sqrt{(4 + 289)} = \sqrt{293}$

Unit vector $= \frac{-2}{\sqrt{293}}\bar{i} - \frac{17}{\sqrt{293}}\bar{j}$

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Answer: $(-2/\sqrt{293})\mathbf{i} - (17/\sqrt{293})\mathbf{j}$

8. If $\mathbf{a} = 3\mathbf{i} - \mathbf{j}$, $\mathbf{b} = -2\mathbf{i} + 4\mathbf{j}$ and $\mathbf{c} = \mathbf{i} + 2\mathbf{j}$, then find a unit vector parallel to $3\mathbf{a} - 2\mathbf{c} + 4\mathbf{b}$.

$3\mathbf{a} - 2\mathbf{c} + 4\mathbf{b} = 3(3\bar{i} - \bar{j}) - 2(\bar{i} + 2\bar{j}) + 4(-2\bar{i} + 4\bar{j})$

$= (9 - 2 - 8)\bar{i} + (-3 - 4 + 16)\bar{j}$

$= -\bar{i} + 9\bar{j}$

$|3\mathbf{a} - 2\mathbf{c} + 4\mathbf{b}| = \sqrt{(1 + 81)} = \sqrt{82}$

Unit vector $= -\frac{1}{\sqrt{82}}\,\bar{i} + \frac{9}{\sqrt{82}}\,\bar{j}$

Answer: $(-1/\sqrt{82})\mathbf{i} + (9/\sqrt{82})\mathbf{j}$

9. Which of the following vectors are parallel?

Two vectors are parallel when one is a constant multiple of the other.

Checking for parallel vectors

Part	Vectors	Ratio	Answer
(i)	$6\mathbf{i} + \mathbf{j}$ and $12\mathbf{i} + 2\mathbf{j}$	$12/6 = 2$, $2/1 = 2$	Parallel
(ii)	$-2\mathbf{i} + 3\mathbf{j}$ and $6\mathbf{i} - 9\mathbf{j}$	$6/(-2) = -3$, $(-9)/3 = -3$	Parallel
(iii)	$5\mathbf{i} - 4\mathbf{j}$ and $6\mathbf{i} - 3\mathbf{j}$	$6/5 \neq (-3)/(-4)$	Not parallel
(iv)	$3\mathbf{i} - 7\mathbf{j}$ and $6\mathbf{i} - 14\mathbf{j}$	$6/3 = 2$, $(-14)/(-7) = 2$	Parallel

Answer: (i), (ii) and (iv) are parallel; (iii) is not parallel

10. Find a vector thrice the length of $3\mathbf{i} - 2\mathbf{j}$, but opposite in direction.

Required vector $= -3(3\bar{i} - 2\bar{j})$

$= -9\bar{i} + 6\bar{j}$

Its magnitude is $3 \times \sqrt{(9 + 4)} = 3\sqrt{13}$ and the sign is opposite, so the direction is opposite too.

Answer: $-9\mathbf{i} + 6\mathbf{j}$

11. Find two vectors that are double in magnitude of $3\mathbf{i} - 5\mathbf{j}$, one in the same direction of it and the other in its opposite direction.

Same direction: $2(3\bar{i} - 5\bar{j}) = 6\bar{i} - 10\bar{j}$

Opposite direction: $-2(3\bar{i} - 5\bar{j}) = -6\bar{i} + 10\bar{j}$

Answer: $6\mathbf{i} - 10\mathbf{j}$ and $-6\mathbf{i} + 10\mathbf{j}$

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